Anindita Sarker Aloka

B.Tech EEE, NIT Trichy | Tiruchirappalli, Tamil Nadu, India
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Education

National Institute of Technology, Trichy — Tiruchirappalli, TN
Bachelor of Technology - Electrical and Electronics Engineering — Expected: 2027
Relevant Coursework: Power Systems, Power Distribution, Control Systems, Analog & Digital Electronics, Converter Topologies, Linear Integrated Circuits, Solar and Wind Energy Systems

Research Experience

Research Intern | IIT Kharagpur — May 2026 - Jul 2026
Supervisor: Dr. Tanmoy Bhattacharya

- Designed and simulated a three-phase 12-pulse Dual Active Bridge converter (Y-Y and Delta-Y bridges, harmonic cancellation) in MATLAB/Simulink, with an outer PI voltage loop and feed-forward current compensation; adapted as the primary side of the group's MMC-based converter.
- Implemented gate pulse generation and discrete control on a Digilent Genesys Virtex-5 FPGA using Xilinx System Generator.
- Ran back-to-back circulating-power hardware tests (592 W, 2 kHz), verified ZVS and efficiency in bidirectional power flow.
- Troubleshot the MMC-based secondary stage of the group's offshore wind converter.

Research Intern | NIT Trichy — Feb 2025 - Present
Supervisor: Dr. Vignesh Kumar

- Simulated a single-phase Dual Active Bridge converter in MATLAB; produced electrical schematics and PCB layouts for the group's hardware prototype.
- Designed, assembled, and tested a signal conditioning circuit PCB for measurement instrumentation, utilizing electrical design calculations for component selection and placement optimization
- Designing conducted EMI filters and PCB trace geometry to mitigate high-frequency switching noise (dv/dt) in prototype power converters.

Leadership & Project Management

Autonomous System Coordinator | EVER NITT — Jan 2024 - Sep 2024

- Built the basic framework for the autonomous system, including drive train, base control board, and worked in hardware design and software integration
- Collaborated with cross-functional team to develop small-scale hybrid vehicle model with integrated autonomous navigation capabilities
- Ensured compatibility between vehicle chassis & autonomous control system through iterative design and testing

Deputy Chief - Graphics Design | Astronomical — Aug 2022 - Sep 2023

- Led and mentored graphics design team, conducting regular training sessions on design principles
- Coordinated design projects with department heads to maintain brand consistency and managed multiple concurrent projects while ensuring quality standards

Technical Projects

Quadrature Oscillator Sine-Cosine Wave Generator — Nov 2025
Mini project (Circuit Design & Simulation)

- Designed and implemented an adjustable-frequency sine/cosine wave generator with voltage control (1 kHz nominal frequency, 2.4 V peak amplitude) on a custom PCB for modular deployment

Multi-Source Solar Charge Controller with MPPT — Nov 2025
Mini project (Circuit Design & Simulation)

- Designed and simulated multi-source charge controller board that automatically selects between solar, alternator, and backup battery inputs while maintaining a 3.7V regulated output using a synchronous buck converter

Custom-Built Interactive BCD to Decimal Conversion Display — Nov 2024
Mini project (PCB Design)

- Designed and implemented a custom PCB for BCD-to-7-segment display converter using the 74LS47 IC

Concrete Speaker Design & Fabrication — 2022

- Designed and fabricated acoustic housing using Fusion 360 3D modeling for enhanced stability
- Assembled electrical components through manual wiring, creating functional audio system

Skills & Certifications

Programming Languages: C, C++
Engineering Software: MATLAB/Simulink, LTSpice, KiCad, EasyEDA, Fusion 360
Hardware & Technical Skills: PCB Design, Schematic Capture, Single-Line Diagrams (SLDs), Electrical Calculations, Circuit Assembly, Xilinx System Generator and FPGA design
Certifications: Introduction to IOT (NPTEL)
Achievements and others:

- International Scholarships: ICCR (Indian Council for Cultural Relations) Scholarship Awardee, 2023
- Translator & Content Manager: Volunteer, UNESCO Bangladesh (Jul 2021 - Sep 2021)

Research Interests

DC mini-grids | Power converters | Renewable energy systems | Battery-aware energy management